

BAIM — Bounded Autonomy Integrity Module

OOF™ Origin Open Foundation™

Independent Methodological Authority

Modul2 BAIM — Bounded Autonomy Integrity Module

Parent Standard: Operational Constraint Integrity Standard (OCNS)

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Operational Reality Standards™

Operational Layer: Cognition Governance Layer

Category: Governance & Enforcement

Subcategory: Bounded Autonomy Integrity

Type: Operational Constraint Integrity Module

Version: 1.0

Status: Canonical · Open Module

Effective Date: 19 May 2026

Compatibility: OOF Methodology OS · Operational Constraint Integrity Standard (OCNS) · Operational Decision Integrity

Standard (ODIS) · Operational Authority Integrity Standard (OAIS) · Operational Boundary Integrity Standard (OBIS) ·

Runtime Integrity Standard (RIS) · Operational Evidence & Auditability Standard (OEAS) · INTEGROS® — Integrity

Standard · Autonomous Runtime Systems · Robotics Infrastructures

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition

Bounded Autonomy Integrity Module (BAIM) defines the structural conditions under which bounded autonomous behavior, runtime autonomy limits, adaptive operational restrictions, and consequence-bearing autonomy states remain materially stable, traceable, and operationally governable across autonomous runtime environments.

A system satisfies BAIM only if:

- bounded autonomous behavior remains materially stable
- runtime autonomy limits preserve governance-valid operational continuity
- adaptive operational restrictions remain coherent
- consequence-bearing autonomy continuity remains traceable
- autonomy expansion does not destabilize operational legitimacy
- A system that preserves runtime execution while bounded autonomy materially destabilizes does not satisfy BAIM.

Module Operational Role

BAIM defines the bounded autonomy layer of OCNS by preserving governance-valid operational autonomy limits across autonomous runtime

environments.

Module Operational Space

BAIM governs the bounded autonomy space of OCNS by preserving runtime autonomy limits, adaptive restriction continuity, operational constraint coherence, and consequence-bearing autonomy stability across runtime systems.

Module Function

- The module applies wherever systems must preserve:
- bounded autonomous behavior
- runtime autonomy limits
- adaptive operational restrictions
- operational constraint legitimacy
- consequence-bearing autonomy continuity

governance-valid operational limits Its function is to ensure that autonomous behavior remains materially bounded strongly enough to preserve governancevalid operational execution across autonomous runtime environments.

Minimum Implementation Framework

1. Define the Bounded Autonomy Object

The organization must define which autonomous operational behaviors require bounded-governance preservation. This may include: autonomous execution systems adaptive runtime agents robotics autonomy infrastructures orchestration autonomy pathways distributed operational autonomy consequence-bearing autonomous actions operational-critical autonomy mechanisms

2. Define Bounded Autonomy Integrity Conditions

The system must define the conditions under which bounded autonomy remains materially stable and operationally aligned. This includes: runtime autonomy continuity adaptive restriction coherence operational constraint legitimacy bounded execution persistence governance-valid autonomy continuity

3. Define Autonomy Expansion Detection Logic

The system must define how materially unstable autonomy conditions or autonomy expansion is identified. This may include: runtime autonomy drift adaptive restriction bypass distributed autonomy divergence bounded execution instability consequence-bearing autonomy escalation

4. Define Operational Response or Governance Logic

The system must define governance logic for materially unstable bounded-autonomy conditions. Governance response may include: autonomy

stabilization adaptive restriction correction distributed autonomy
synchronization bounded execution reconstruction operational review
activation operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of
bounded-autonomy continuity and autonomy-expansion states. A system must
not remain constraint-valid if bounded autonomous behavior materially
destabilizes while systems continue assuming governance-valid
operational limits remain preserved.

Use Case 1 — Autonomous AI Execution

Infrastructure

Scenario

A distributed AI infrastructure continuously performs adaptive
operational execution across orchestration systems and autonomous
runtime environments.

Application

BAIM preserves governance-valid bounded autonomy through runtime
autonomy governance and adaptive restriction stabilization.

Result

The organization gains stronger operational autonomy continuity and
reduced hidden autonomy expansion across autonomous runtime systems.

Use Case 2 — Robotics Autonomous

Coordination Environment

Scenario

A robotics infrastructure continuously performs adaptive autonomous
behavior across distributed operational systems and realtime
environments.

Application

BAIM preserves governance-valid bounded execution through adaptive
autonomy governance and distributed operational restriction
synchronization.

Result

The environment gains stronger operational autonomy legitimacy and
reduced runtime autonomy instability across autonomous operational

ecosystems.

Canonical Closing Statement

If autonomous behavior cannot remain materially bounded across autonomous runtime environments, systems may preserve operational execution while governance-valid operational limits progressively destabilize across distributed autonomy layers.

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